Networking Module 6

S Vimal

1. Capture and analyze ARP packets using Wireshark. Inspect the ARP request and reply frames when your device attempts to find the router's MAC address. Discuss the importance of ARP in packet forwarding.

For this experiment I used Wireshark on my VirtualBox's Ubuntu machine. Here

The default gateway of my ubuntu: 10.0.2.2

The ip address of my ubuntu: 10.0.2.15

And I pinged an IP 192.168.1.1. Here is the output I got while capturing ARP packets.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	PCSSystemtec_f5:68:…	Broadcast	ARP	42	Who has 10.0.2.2? Tell 10.0.2.15
2	0.000574585	52:54:00:12:35:02	PCSSystemtec_f5:68:…	ARP	60	10.0.2.2 is at 52:54:00:12:35:02

<u>Analysing packet 1:</u>

```
▸ Frame 1: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface enp0s3, id 0
▸ Ethernet II, Src: PCSSystemtec_f5:68:17 (08:00:27:f5:68:17), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▾ Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: PCSSystemtec_f5:68:17 (08:00:27:f5:68:17)
    Sender IP address: 10.0.2.15
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 10.0.2.2
```

It is a broadcast type packet,a request to know which MAC has the IP 10.0.2.2.

Sender address: 10.0.2.15 (Ubuntu VM's IP)

Target IP: 10.0.2.2 (Gateway)

Target MAC: Unknown. This information is what we need.

Analysing packet 2:

```
▸ Frame 2: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface enp0s3, id 0
▸ Ethernet II, Src: 52:54:00:12:35:02 (52:54:00:12:35:02), Dst: PCSSystemtec_f5:68:17 (08:00:27:f5:68:17)
▾ Address Resolution Protocol (reply)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    Sender MAC address: 52:54:00:12:35:02 (52:54:00:12:35:02)
    Sender IP address: 10.0.2.2
    Target MAC address: PCSSystemtec_f5:68:17 (08:00:27:f5:68:17)
    Target IP address: 10.0.2.15

"SSS" is neither a field nor a protocol name.
```

This is not a broadcast type message, it is a reply message.

Sender IP: 10.0.2.2 (Gateway)

Sender MAC: 52:54:00:12:35:02, knowing this MAC was the purpose of these ARP packets.

Target : Ubuntu VM

ARP in Packet Forwarding:

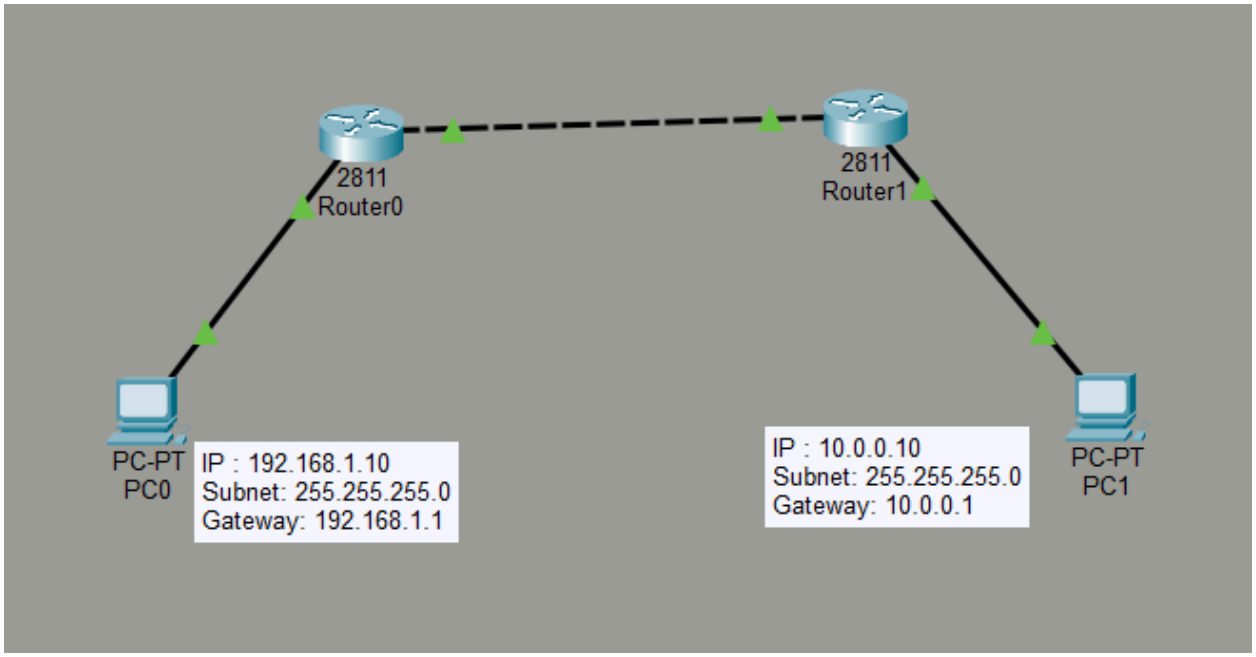
ARP (Address Resolution Protocol) is fundamental to how packets are forwarded across a local network. Every time a device wants to send data, it needs both the destination IP address and the destination MAC address to construct a proper Ethernet frame. While the IP address is known from the application layer, the MAC address must be discovered dynamically using ARP. When a device wants to reach a destination outside its subnet, it does not send ARP for the remote IP directly, instead it sends an ARP request for the default gateway's MAC address, since all traffic leaving the local network must pass through the router first.

When the ARP request is broadcast across the network, every device receives it but only the router responds with its MAC address. The requesting device stores this mapping in its ARP cache and uses it to forward all outgoing packets to the router. Without this ARP resolution, the device cannot construct the Ethernet frame and no packet can leave the local network regardless of whether the IP routing is correctly configured. This makes ARP an essential prerequisite for packet forwarding at the data link layer.

2. Manually configure static routes on a router to direct packets to different subnets.

Use the ip route command and verify connectivity using ping and traceroute.

Cisco Setup



IP and routing configuration on router 0

```
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 172.16.0.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
exit
Router(config)#ip route 10.0.0.0 255.255.255.0 172.16.0.2
```

Ip route command is used to route subnet of router0, it just tells in order to route packets to traffic 10.0.0.0/24, the next hop is 172.16.0.2 router1's interface with router0

IP and routing configuration on router 1

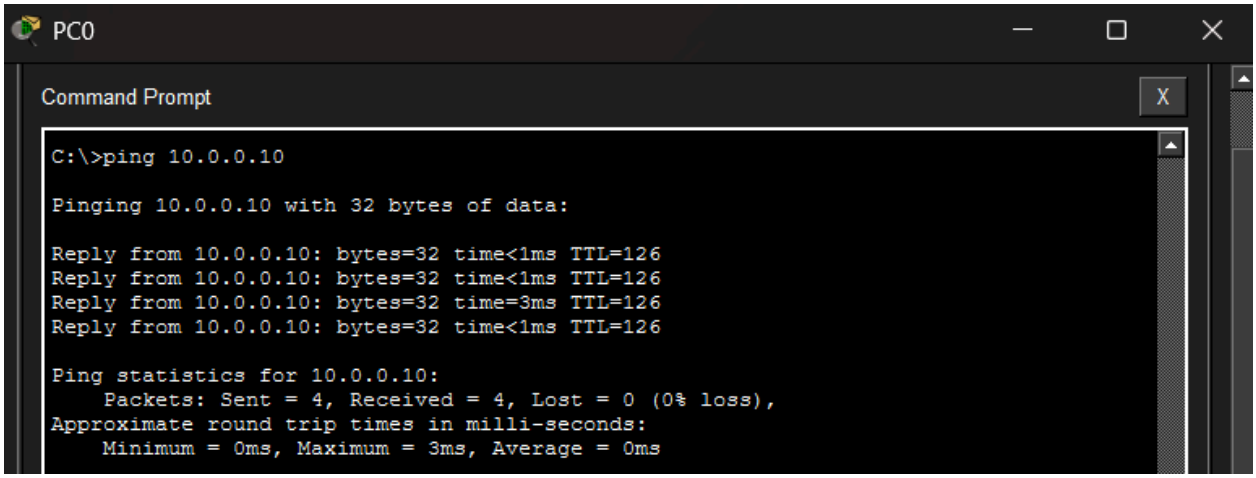
```
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 172.16.0.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 172.16.0.2 255.255.255.0
% 172.16.0.0 overlaps with FastEthernet0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#no shutdown


Router(config)#ip route 192.168.1.0 255.255.255.0 172.16.0.1
```

Similarly iproute is used here to tell router1 that to reach 192.168.1.0/24 the next hop is 172.16.0.1, router0's interface with router1

Verifying using show ip route

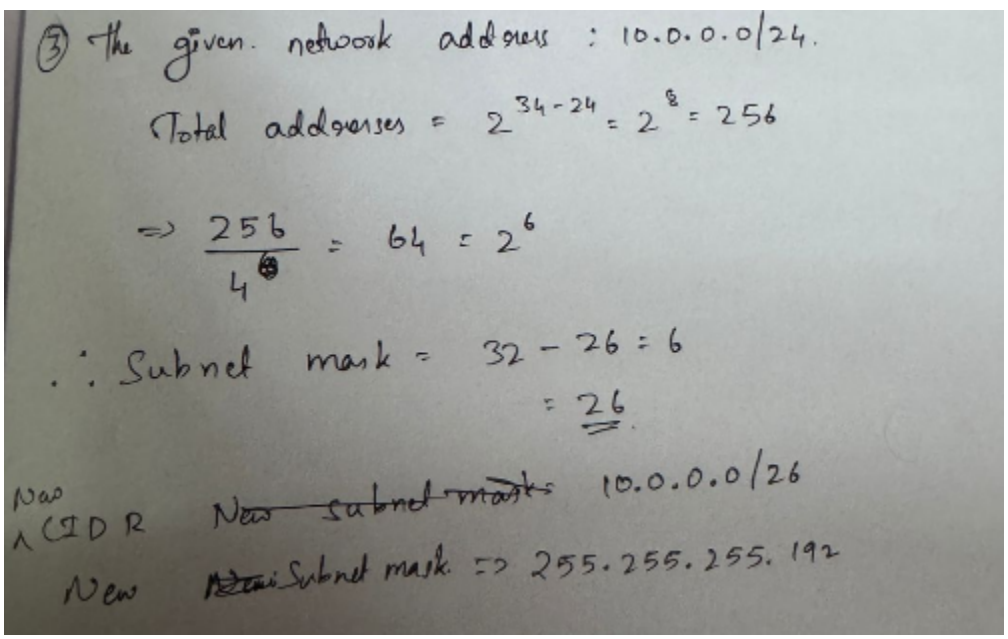
```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 1 subnets
S       10.0.0.0/24 [1/0] via 172.16.0.2
     172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/24 is directly connected, FastEthernet0/1
L       172.16.0.1/32 is directly connected, FastEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, FastEthernet0/0
L       192.168.1.1/32 is directly connected, FastEthernet0/0
```

Verifying by pinging PC1 from PC0



3. Given a network address of 10.0.0.0/24, divide it into 4 equal subnets.

- **Calculate the new subnet mask.**
- **Determine the valid host range for each subnet.**
- **Assign IP addresses to devices in Packet Tracer and verify connectivity.**



③ the given network address : 10.0.0.0/24.

Total addresses = $2^{34-24} = 2^8 = 256$

$\Rightarrow \frac{256}{4} = 64 = 2^6$

∴ Subnet mask = 32 − 26 = 6

= 26.

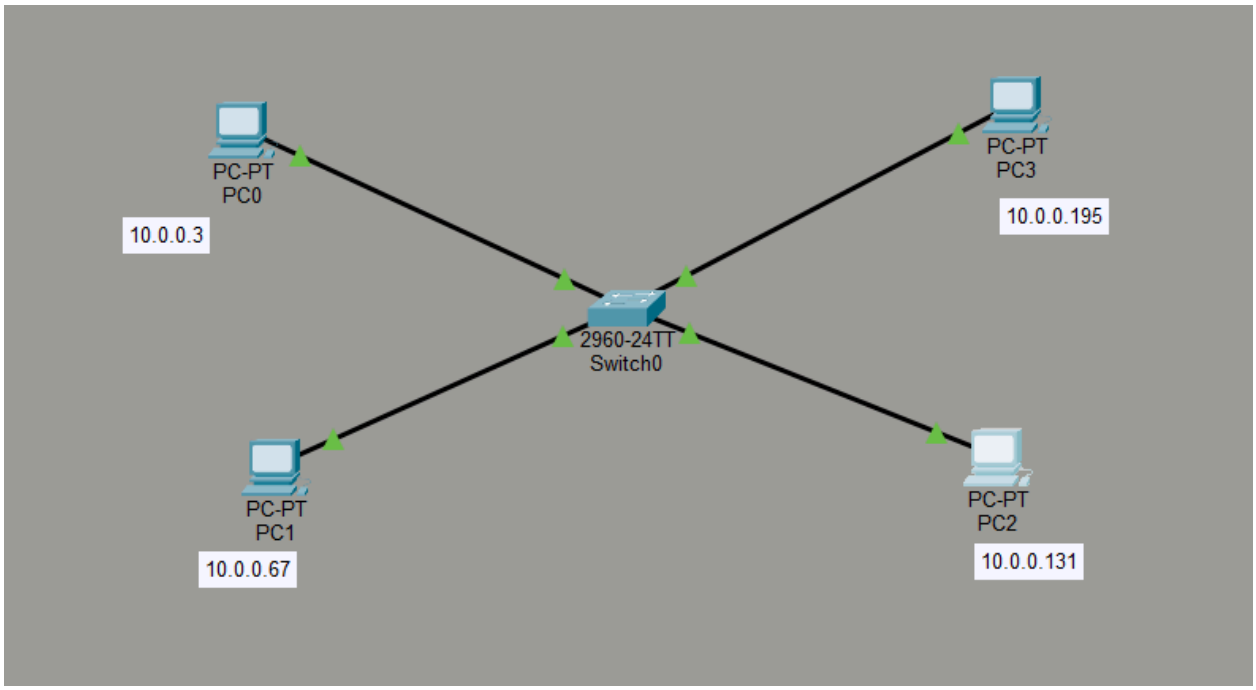
New CIDR ~~New subnet mask~~ 10.0.0.0/26

New ~~New~~ Subnet mask => 255.255.255.192

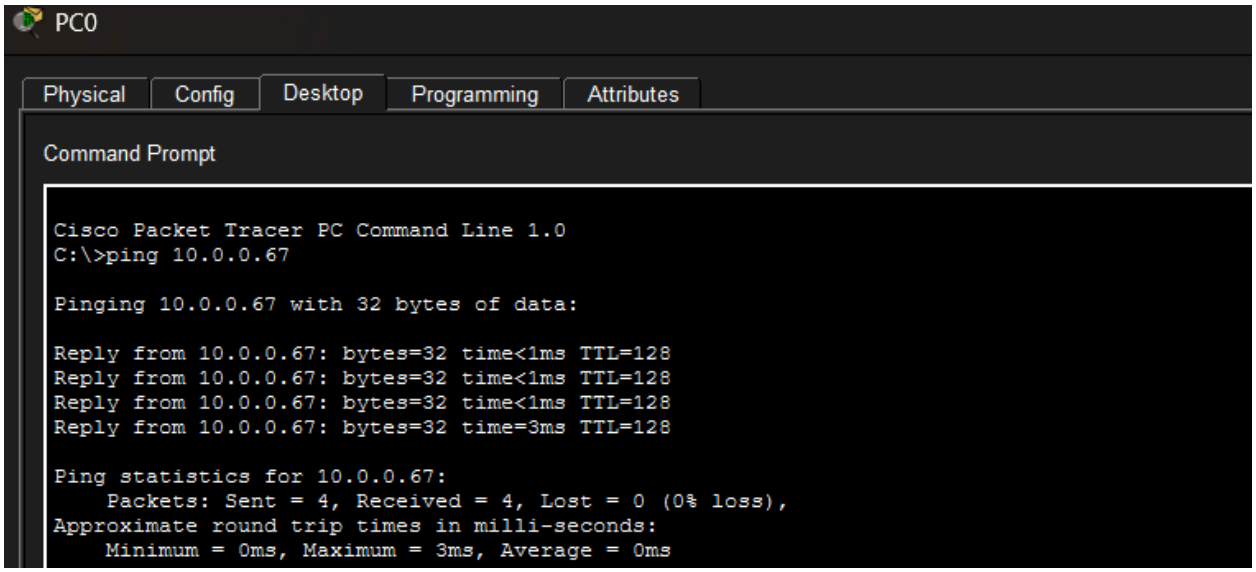
Subnet	Host Range	Network Address	Broadcast Address
Subnet 1	10.0.0.1 to 10.0.0.62	10.0.0.0/26	10.0.0.63
Subnet 2	10.0.0.65 to 10.0.0.126	10.0.0.64/26	10.0.0.127
Subnet 3	10.0.0.129 to 10.0.0.190	10.0.0.128/26	10.0.0.191
Subnet 4	10.0.0.193 to 10.0.0.254	10.0.0.192/26	10.0.0.255

Configuring in Cisco Packet Tracer

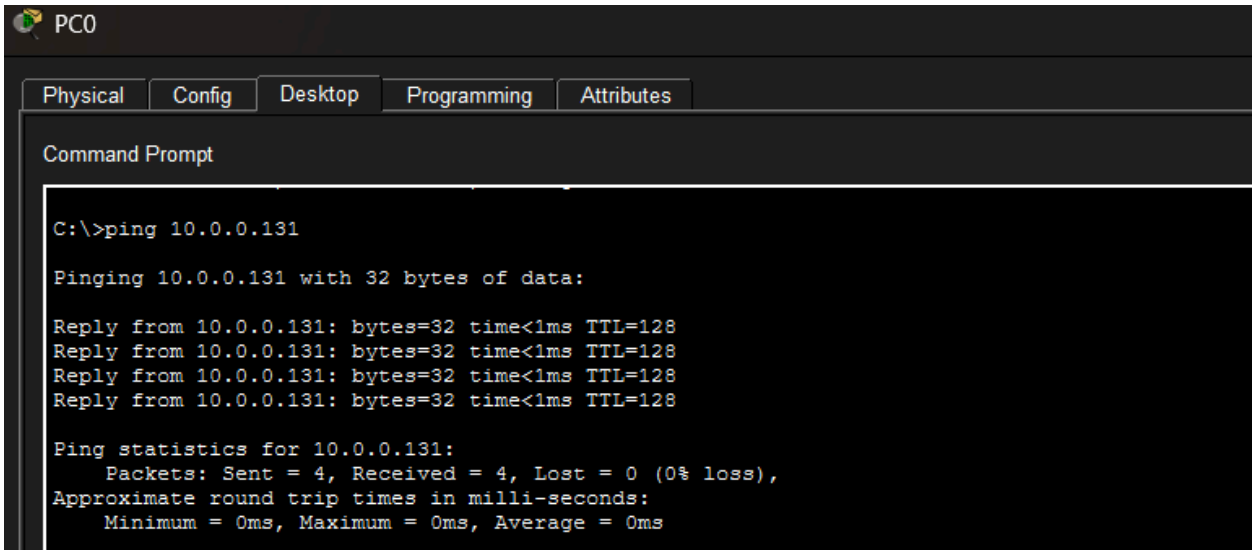
Subnet	IP used
Subnet 1	10.0.0.3
Subnet 2	10.0.0.67
Subnet 3	10.0.0.131
Subnet 4	10.0.0.195



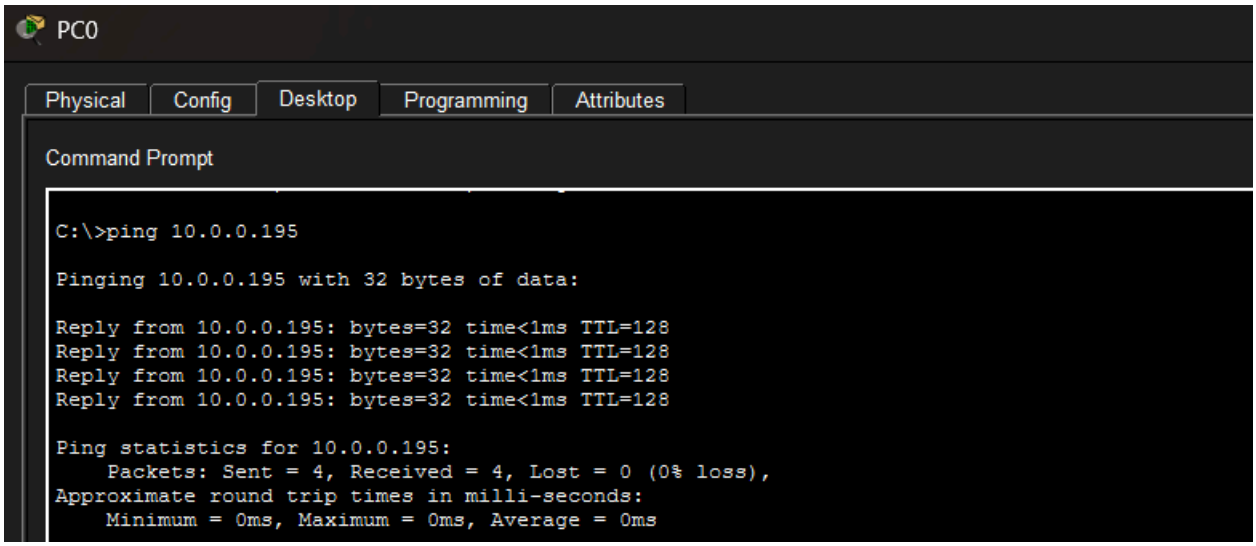
PING PC0 to PC1



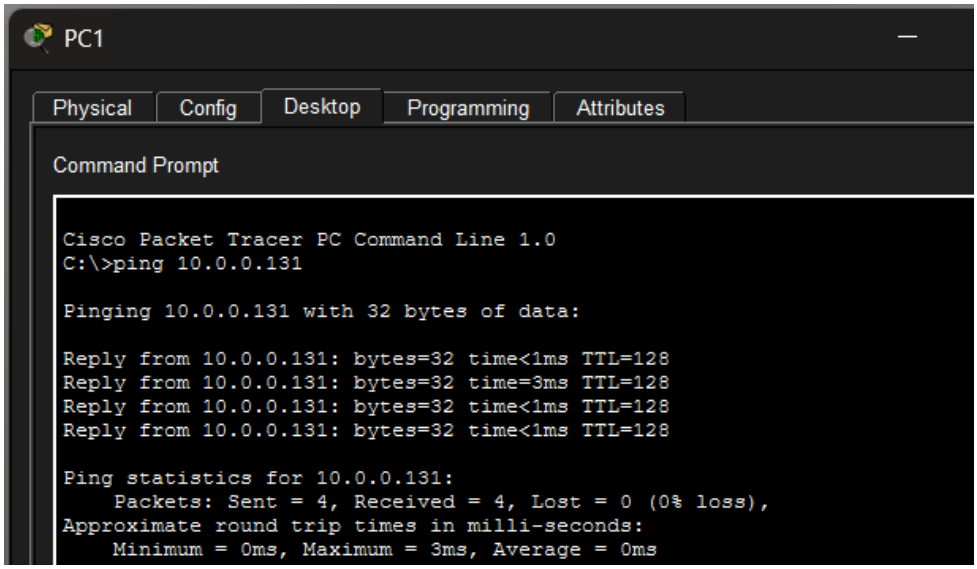
PING PC0 to PC2



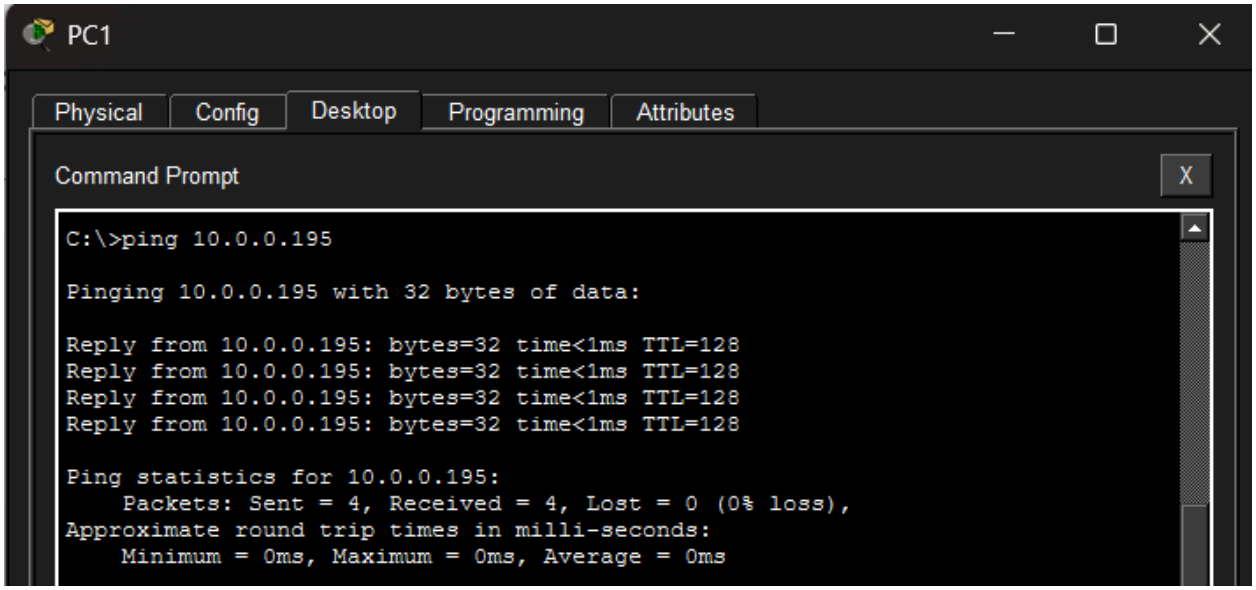
PING PC0 to PC3



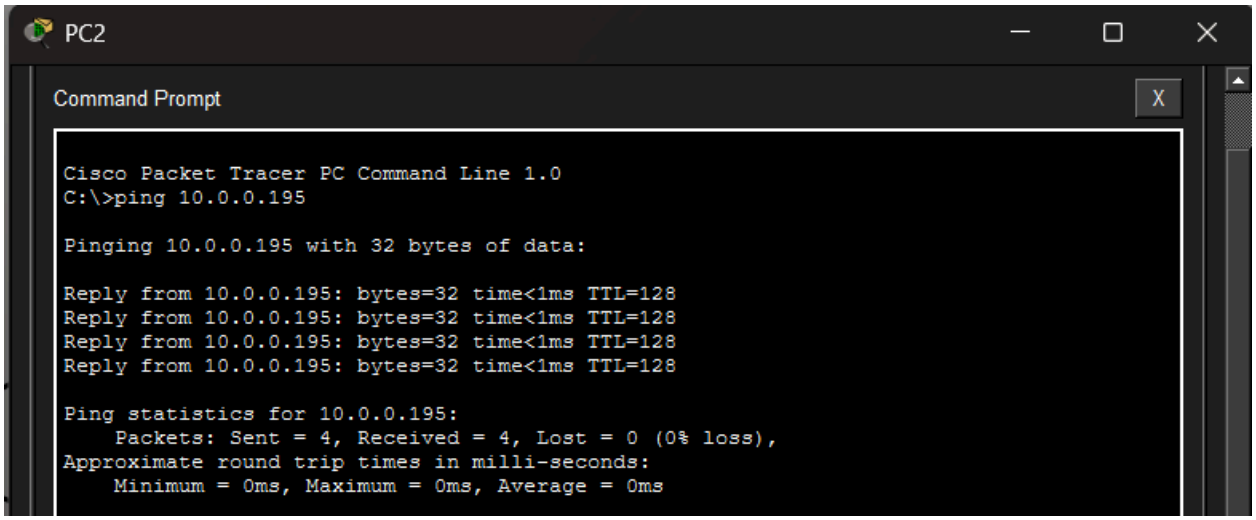
PING PC1 to PC2



PING PC1 to PC3



PING PC2 to PC3



4. You are given three IP addresses: 192.168.10.5, 172.20.15.1 , and 8.8.8.8.

Identify the class of each IP address.

Determine if it is private or public.

Explain how NAT would handle a private IP when accessing the internet.

IP Address	Class	Reason	Public/Private	Reason
192.168.10.5	Class C	192.0.0.0 to 223.255.255.255	Private	192.168.0.0 to 192.168.255.255
172.20.15.1	Class B	128.0.0.0 to 191.255.255.255	Private	172.16.0.0 to 172.31.255.255
8.8.8.8	Class A	1.0.0.0 to 126.255.255.255	Public	No private ranges

<u>How NAT Handles a Private IP</u>

Consider a situation, where a PC which in a network having local IP sends packet to the internet, and these are steps that will take place in the router, since local IP cannot be routed.

Step 1: PC Sends Traffic A device with private IP 192.168.10.5 generates a packet destined for an external server. Since private IPs are not routable on the internet, it must pass through the router first.

Step 2: Router Intercepts the Packet The router receives the packet on its inside interface and identifies the source as a private IP that requires translation before being forwarded.

Step 3: NAT Translation Occurs The router replaces the private source IP 192.168.10.5 with its own public IP and assigns a unique port number to distinguish sessions from multiple internal devices.

Step 4: NAT Table Updated The router records the mapping between the private IP and port, and the translated public IP and port in its NAT translation table for tracking return traffic.

Step 5: Packet Sent to Internet The translated packet leaves through the outside interface. The external server sees only the router's public IP and has no knowledge of the internal private address.

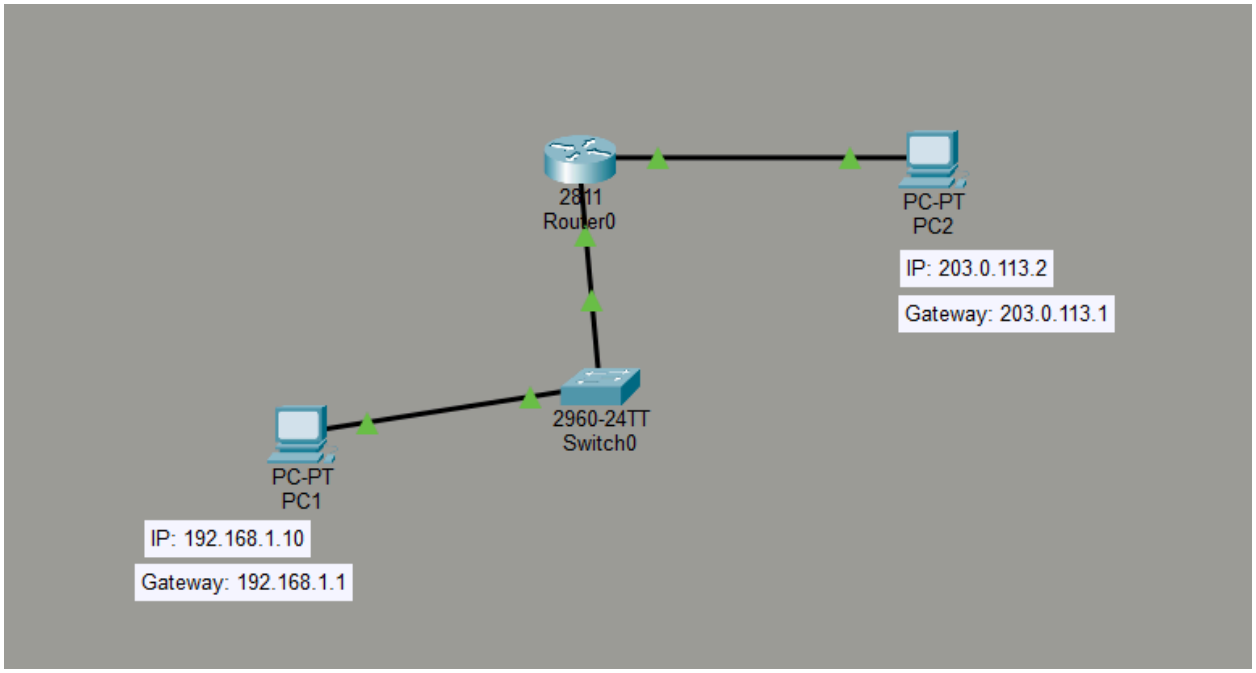
Step 6: Reply Returns The external server sends its reply back to the router's public IP. The router checks its NAT table, identifies the correct internal device, replaces the public destination IP back to the private IP 192.168.10.5, and delivers the packet to the original device.

5. In Cisco Packet Tracer, configure NAT on a router to allow internal devices

(192.168.1 .x) to access the internet. Test connectivity by pinging an external public IP.

Capture the traffic in Wireshark and analyze the source IP before and after NAT Translation.

Cisco Packet Tracer Config



Configuring NAT on router inside

```
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shutdown
```

Configuring NAT on router outside

```
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 203.0.113.1 255.255.255.0
Router(config-if)#ip nat outside
Router(config-if)#no shutdown
```

Configuring Access List

```
exit
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface FastEthernet0/1 overload
```

Verifying with ping

PC1

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 203.0.113.2

Pinging 203.0.113.2 with 32 bytes of data:

Request timed out.
Reply from 203.0.113.2: bytes=32 time<1ms TTL=127
Reply from 203.0.113.2: bytes=32 time<1ms TTL=127
Reply from 203.0.113.2: bytes=32 time=3ms TTL=127

Ping statistics for 203.0.113.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

Verifying by running NAT Translations

```
Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 203.0.113.1:5     192.168.1.10:5     203.0.113.2:5      203.0.113.2:5
icmp 203.0.113.1:6     192.168.1.10:6     203.0.113.2:6      203.0.113.2:6
icmp 203.0.113.1:7     192.168.1.10:7     203.0.113.2:7      203.0.113.2:7
icmp 203.0.113.1:8     192.168.1.10:8     203.0.113.2:8      203.0.113.2:8
```

This means the intended NAT Translation has happened, the IP 192.168.1.10:5, represents the IP of inside the local network before reaching the router, and the sequence :5, just for tracking the ICMP packet. And it is mapped to inside global 203.0.113.1, which is the IP interface of the router from the outside, all packets meant to reach this network will have this address. And after reaching the router, the address will translate to the local address and packets move to the intended device.

Verifying by running NAT Statistics

```
Router#show ip nat statistics
Total translations: 4 (0 static, 4 dynamic, 4 extended)
Outside Interfaces: FastEthernet0/1
Inside Interfaces: FastEthernet0/0
Hits: 7  Misses: 8
Expired translations: 4
Dynamic mappings:
```

This verifies while using the ping command 4 times the translations have been done since we sent out 4 packets for ping